

WIPRO CAPSTONE PROJECT

**[PROJECT NAME: BIGBASKET WEB APPLICATION
AUTOMATION TESTING]**

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MODULE NAME: Fashion

DOMAIN NAME: Selenium with Python

Synopsis

PROJECT DESCRIPTION

SELENIUM WITH PYTEST

Introduction

Automation testing is used to improve software quality by reducing manual effort and increasing testing efficiency. Selenium WebDriver is an open-source automation testing tool used to automate web applications across different browsers. In this project, Selenium is integrated with the Pytest framework using Python programming language.

Pytest is a testing framework used for executing test cases, managing assertions, generating reports, and improving test organization. Selenium with Pytest helps in building scalable and reusable automation frameworks.

The framework automates various functionalities of the BigBasket web application such as:

1. Login validation
2. Product search
3. Product filtering
4. Add to cart functionality
5. Cart validation
6. Checkout process

The framework follows the Page Object Model (POM) design pattern where locators, page methods, and test logic are separated for better maintainability and scalability.

The project also integrates:

1. Allure Reports
2. Screenshot capturing
3. Logging
4. Excel Data-Driven Testing

to generate professional automation reports and improve debugging.

Advantages

1. Reduces manual testing effort
2. Supports reusable automation scripts

ARCHITECTURE

SELENIUM WITH PYTEST – FRAMEWORK ARCHITECTURE

Project_BigBasket

```
|  
|├── .venv/  
|├── config/  
|   └── config.properties  
|├── data/  
|   └── test_data.csv  
|├── logs/  
|   └── automation.log  
|├── pages/  
|   ├── __init__.py  
|   ├── basepage.py  
|   ├── homepage.py  
|   └── loginpage.py  
|├── reports/  
|   ├── allure-report/  
|   ├── allure-results/  
|   └── screenshots/
```

```
└─ tests/
|   └─ __init__.py
|   └─ test_E2E.py
|   └─ tests_negative.py
|   └─ tests_positive.py
└─ utils/
|   └─ __init__.py
|   └─ config_reader.py
|   └─ csv_reader.py
|   └─ excel_reader.py
|   └─ logger.py
|   └─ screenshot_util.py
└─ conftest.py
└─ main.py
└─ pytest.ini
└─ requirements.txt
```

BDD WITH BEHAVE

Introduction

Behavior Driven Development (BDD) is a software testing methodology where test scenarios are written in simple English language using Gherkin syntax. Behave is the Python BDD framework used in this project along with Selenium WebDriver.

BDD improves collaboration between developers, testers, and business stakeholders by making test cases easy to understand and readable.

Feature files are written using:

1. Given
2. When
3. Then

keywords.

Example

Scenario: Add fashion Product To Cart

Given user is logged into BigBasket

When the user clicks on fashion from the homepage

And the user adds Fashion product to cart

Then the Fashion product should be added successfully

The framework automates:

1. Positive test scenarios
2. Negative test scenarios
3. End-to-End testing scenarios

using reusable step definitions and Page Object Model design pattern.

The project also includes:

1. Allure Reporting
2. Screenshot attachments
3. Logging functionality
4. Excel-based data handling

ARCHITECTURE

BDD WITH BEHAVE – FRAMEWORK ARCHITECTURE

BDD_Project_BigBasket

```
|
|├── .venv/
|├── config/
|   └── config.ini
|├── data/
|   └── test_data.csv
|├── features/
|   ├── steps/
|   │   ├── E2E_steps.py
|   │   ├── negative_steps.py
|   │   └── positive_steps.py
|   ├── E2E.feature
|   ├── environment.py
|   ├── negative.feature
|   └── positive.feature
|├── logs/
|   └── automation.log
|├── pages/
|   ├── basepage.py
|   ├── homepage.py
|   └── loginpage.py
```

- └─ reports/
 - └─ allure-report
 - └─ allure-results
 - └─ screenshots
- └─ utils/
 - | └─ action_utils.py
 - | └─ browser_utils.py
 - | └─ config_reader.py
 - | └─ csv_reader.py
 - | └─ logger.py
 - | └─ screenshot_util.py
 - | └─ wait_utils.py
- └─ behave.ini
- └─ main.py
- └─ msedgedriver.exe
- └─ requirements.txt
- └─ runtests.py

TEST STRATEGY

Selenium Functional Testing

Objective

The objective of Selenium Functional Testing is to validate the core functionalities of the BigBasket web application using Selenium WebDriver with Python and Pytest framework.

What We Have Done

The following functional test scenarios were automated:

Positive Test Cases

1. Login functionality validation
2. Tea product search
3. Tea product filtering
4. Add Tea product to cart
5. Delete Tea product from cart
6. Sort Ghee products by price
7. Cart validation
8. Checkout navigation

Negative Test Cases

1. Invalid mobile number validation
2. Invalid OTP validation
3. Invalid Tea product search
4. Invalid card details validation

Framework Design

The Selenium framework was developed using:

1. Python
2. Selenium WebDriver
3. Pytest Framework
4. Page Object Model (POM)

BDD Functional Testing

Objective

The objective of BDD Functional Testing is to validate application behavior using simple English-readable test scenarios with Behave framework.

What We Have Done

BDD scenarios were created for:

1. Login validation
2. Product search validation
3. Product filter validation
4. Cart functionality
5. Product deletion
6. Checkout process
7. Payment validation
8. End-to-End shopping flow

BDD Implementation

The framework was developed using:

1. Behave BDD Framework
2. Gherkin Feature Files
3. Step Definitions
4. Selenium WebDriver
5. Page Object Model

End-To-End Flow Automated

1. User login
2. Add Fashion product
3. Add Fashion brand product
4. Navigate to cart
5. proceed to checkout
6. Validate invalid payment card

TOOLS USED

Tool / Technology	Purpose
Python	Programming Language
Selenium WebDriver	Web Automation
Pytest	Test Execution Framework
Behave	BDD Framework
Allure Reports	HTML Reporting
OpenPyXL	Excel Data Handling
WebDriver Manager	Browser Driver Management
Chrome Browser	Test Execution
PyCharm	IDE
GitHub	Version Control

TEST ENVIRONMENT

Operating System Used

Windows 11

IDE USED

PyCharm Community Edition

TOOLS USED

The following tools and technologies were used in the project:

1. Python Selenium WebDriver
2. Pytest Framework
3. Behave BDD Framework
4. Allure Reports
5. OpenPyXL
6. WebDriver Manager
7. Chrome Browser
8. GitHub

requirements.txt Includes

1. selenium behave
2. pytest
3. allure-behave
4. openpyxl
5. webdriver-manager

FRAMEWORK ARCHITECTURE

The project follows a Hybrid Automation Framework architecture combining:

Selenium Functional Framework

BDD Framework using Behave

1. Data-Driven Testing
2. Page Object Model (POM)

Framework Components

- 1.Feature Files
2. Step Definition Files
3. Page Classes
4. Locator Classes
5. Utility Files
6. Test Data Files
7. Reports
8. Logs

ALLURE REPORTS

Allure Reports are integrated for advanced HTML reporting.

Features Of Allure Reports

1. Pass/Fail Status
2. Step Execution Details
3. Screenshot Attachments
4. Error Logs
5. Failure Analysis
6. Execution Summary

Reports are automatically generated after test execution.

PAGE OBJECT MODEL (POM)

The framework follows the Page Object Model (POM) design pattern.

In POM:

1. Locators are maintained separately
2. Reusable methods are maintained in page classes
3. Test logic is maintained in step definition files

TEST EXECUTION

End-To-End Testing Module

The End-To-End (E2E) Testing module validates the complete user workflow of the BigBasket web application from login to checkout process. The main objective of E2E testing is to ensure that all integrated modules of the application work correctly together without failures.

In this project, the End-To-End testing flow automates the complete shopping process including user login, product selection, cart operations, checkout navigation, and payment validation. The automated E2E flow starts by logging into the application using valid credentials. After successful login, the user searches and adds Tea and Ghee products to the cart, navigates to the cart page, and proceeds to checkout. Finally, invalid card details are entered to validate payment error handling functionality.

The End-To-End module ensures that:

1. User navigation works correctly
2. Product selection works properly
3. Cart functionality behaves correctly
4. Checkout flow is validated
5. Payment validation messages are displayed properly

This testing improves application reliability and verifies real-time user behavior across multiple modules.

TESTS CONDUCTED

A total of:

1. Positive Test Cases
2. Negative Test Cases
3. End-To-End Test Scenarios

were executed during automation testing.

Positive Test Cases Conducted

1. Login functionality validation
2. Fashion product search
3. Fashion product filtering

4. Add Fashion product to cart
5. Delete Fashion product from cart
6. Sort Fashion products
7. Cart validation
8. Checkout navigation

Negative Test Cases Conducted

1. Invalid mobile number validation
2. Invalid OTP validation
3. Invalid Fashion product search
4. Invalid card details validation

End-To-End Scenario Conducted

1. User login
2. Add Fashion product
3. sort brands product
4. Cart validation
5. Checkout process
6. Invalid payment validation

The automated tests were executed successfully using Selenium WebDriver with Pytest and BDD using Behave framework. Allure Reports were generated for execution status, screenshots, and failure analysis.

TEST OUTPUTS

Outputs Screenshots:

TEST FLOW

Total Tests Executed : 7

Positive Test Cases : 4

Negative Test Cases : 2

End-To-End Scenarios : 1

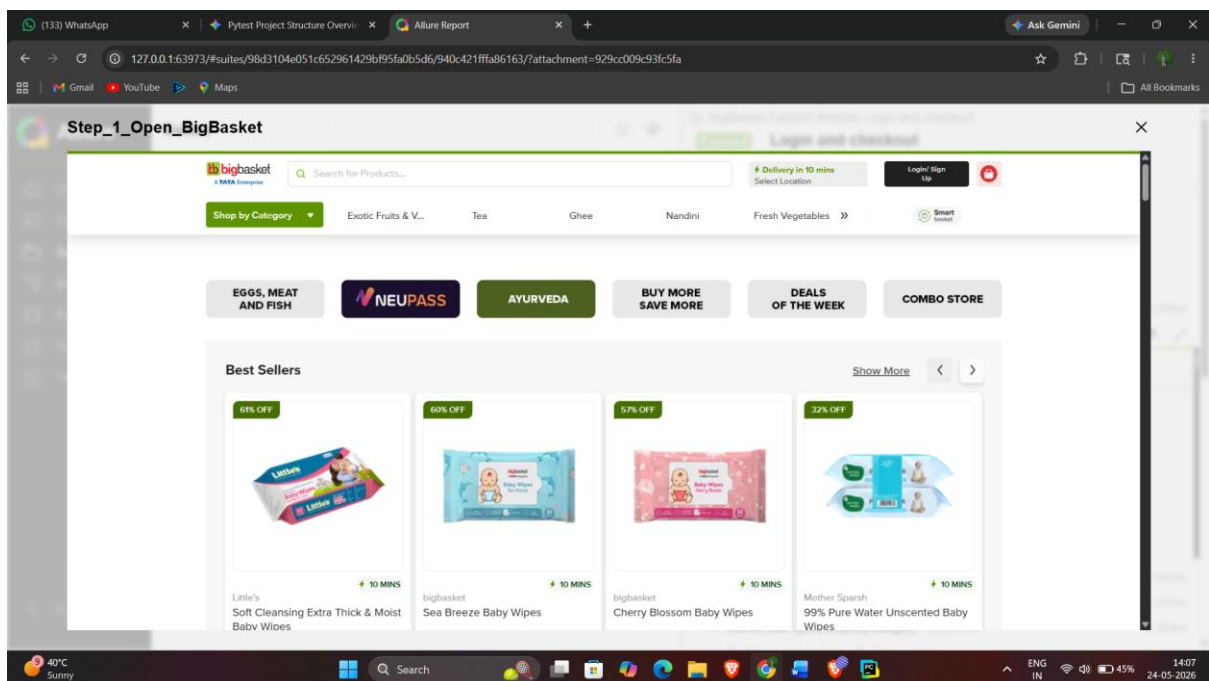
Selenium Functional Testing Flow

Step 1: Launch Browser

1. Chrome browser is launched using Selenium WebDriver.

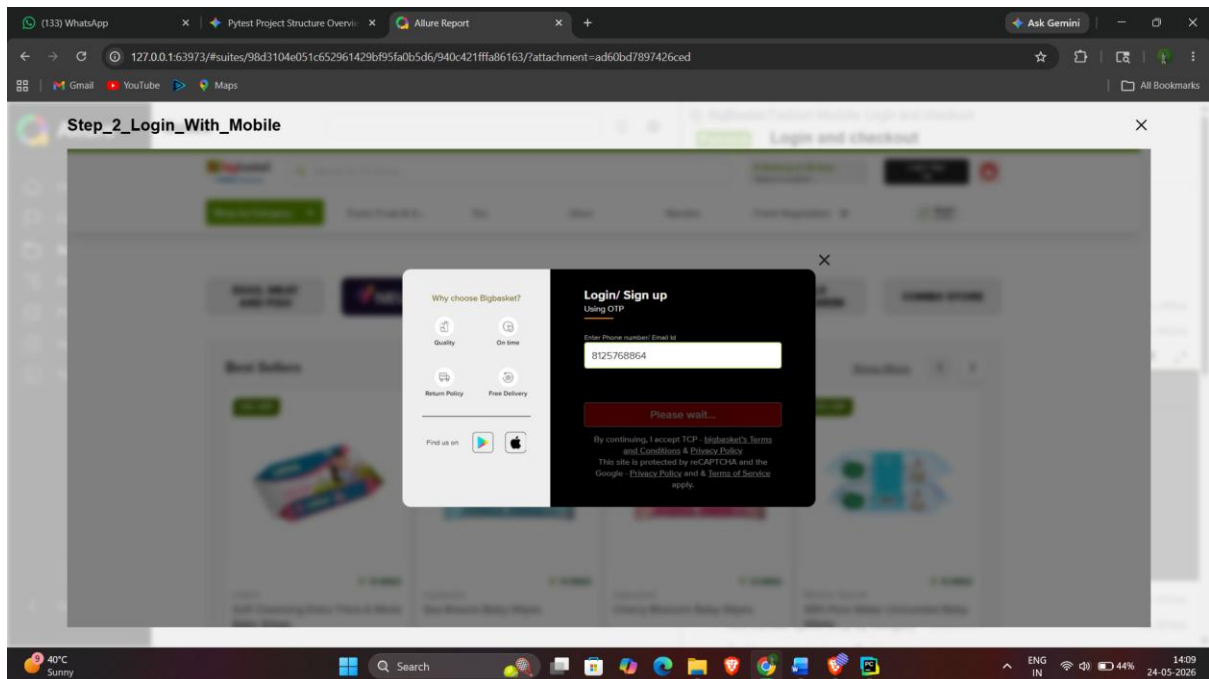
Step 2: Open BigBasket Application

2. BigBasket website URL is opened



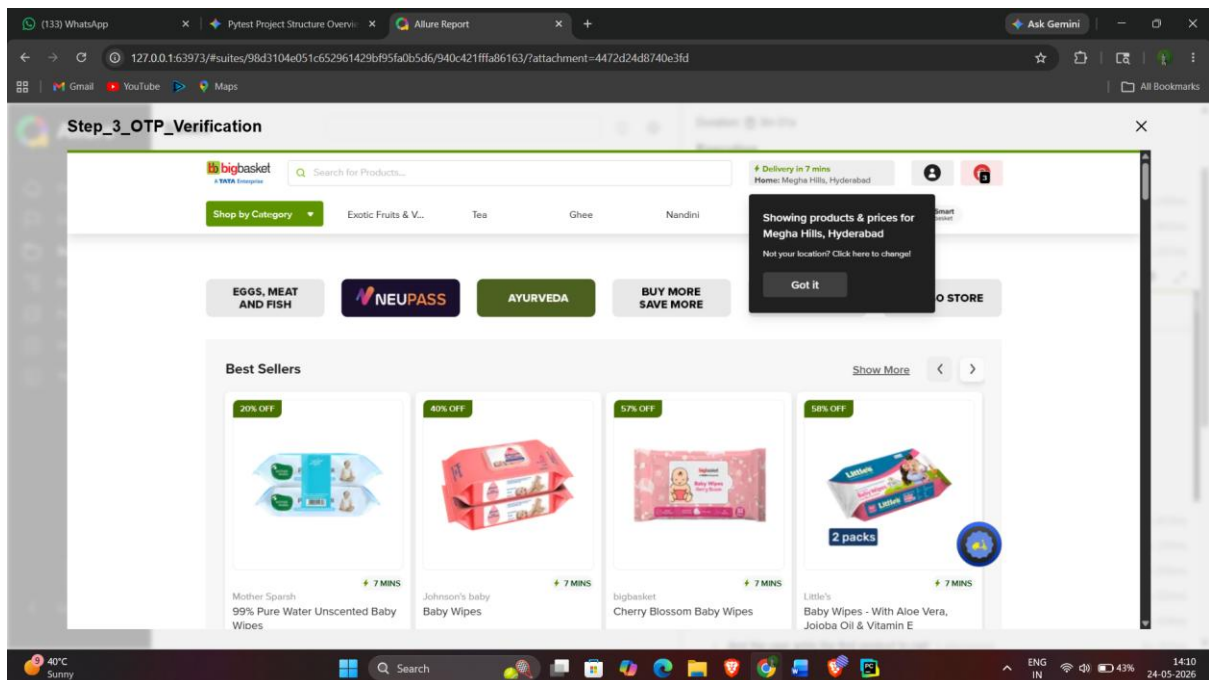
Step 2:-

When the user logs in with valid mobile number



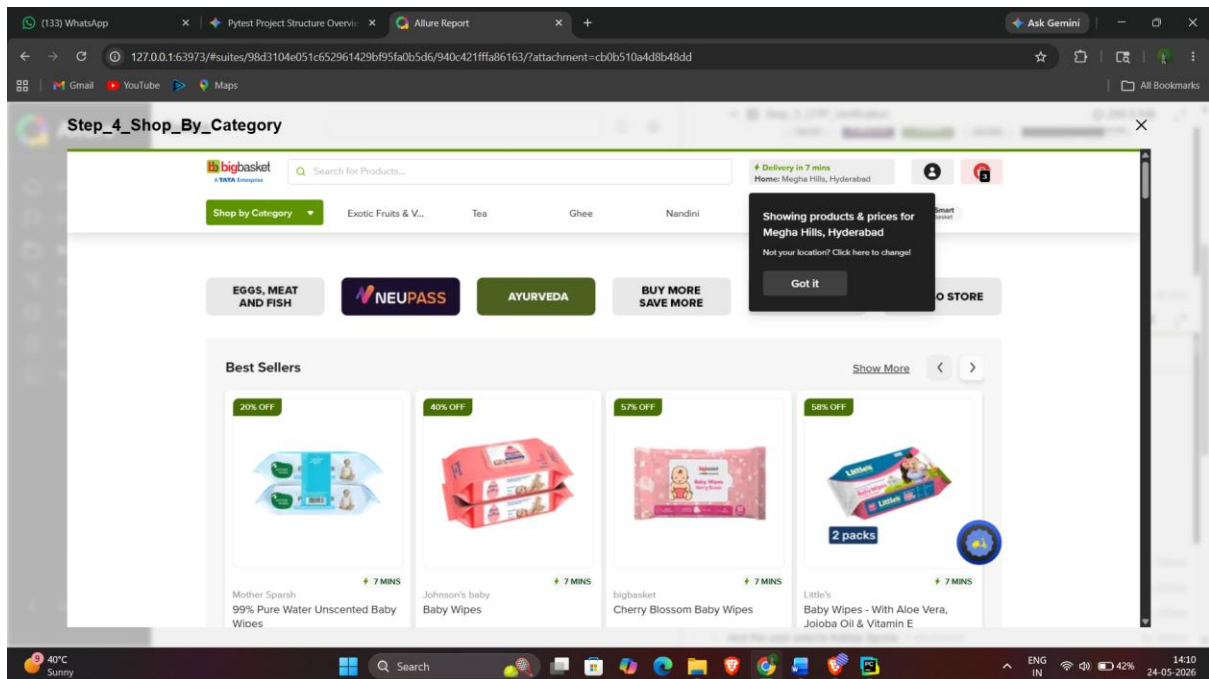
Step 3:-

And the user waits for OTP and verifies login



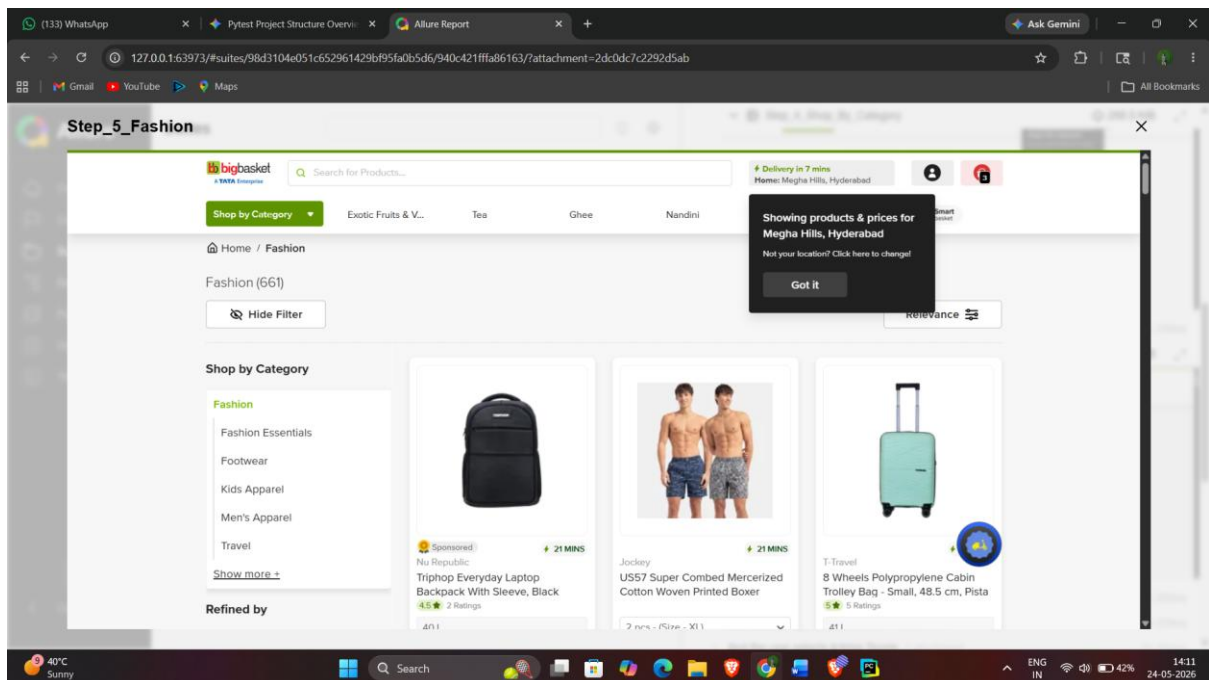
Step 4:-

And the user opens Shop By Category



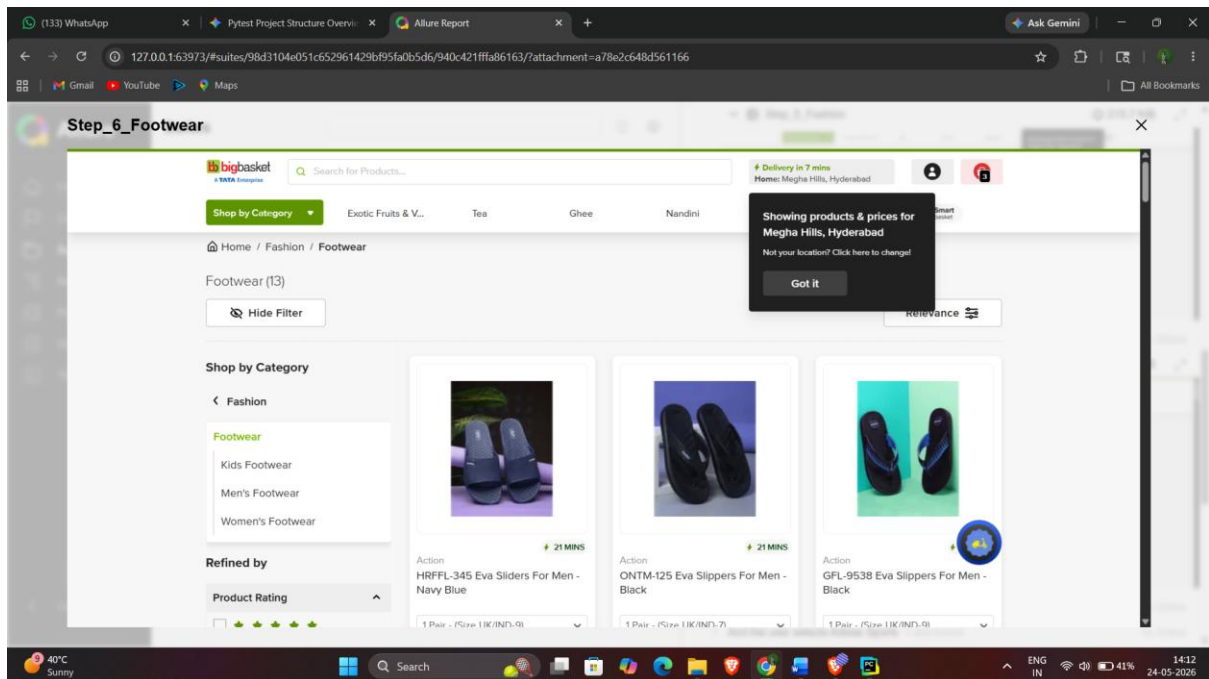
Step 5:-

And the user navigates to Fashion



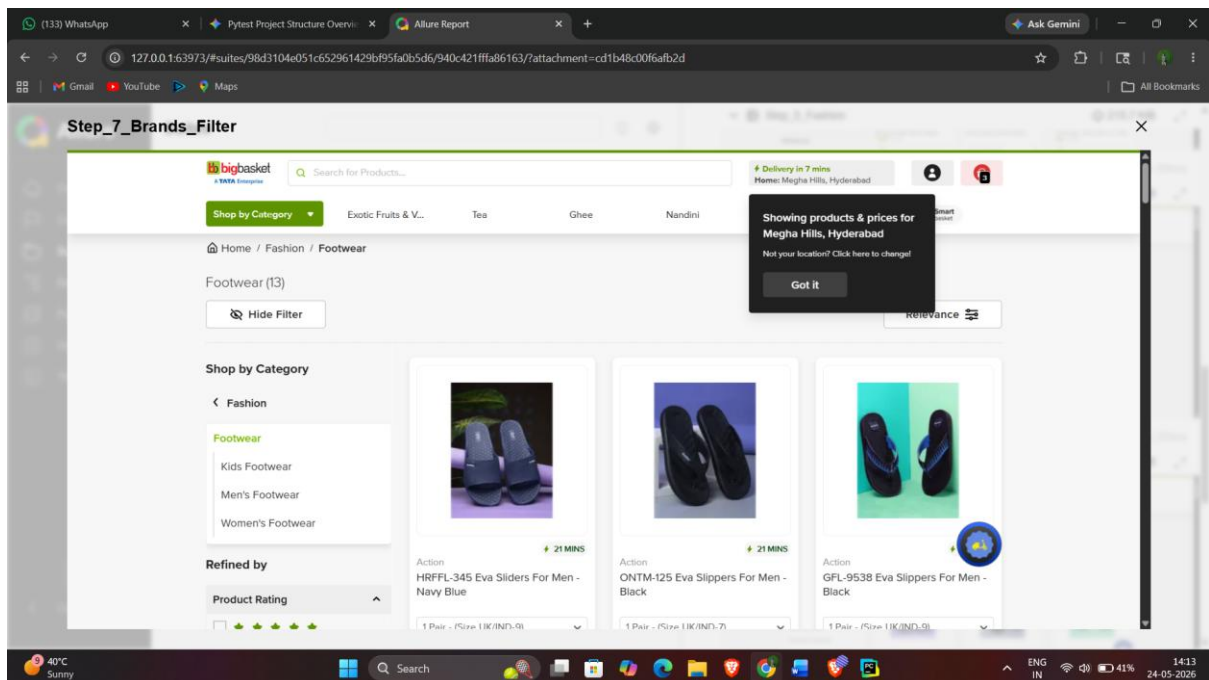
Step 6:-

And the user navigates to Footwear



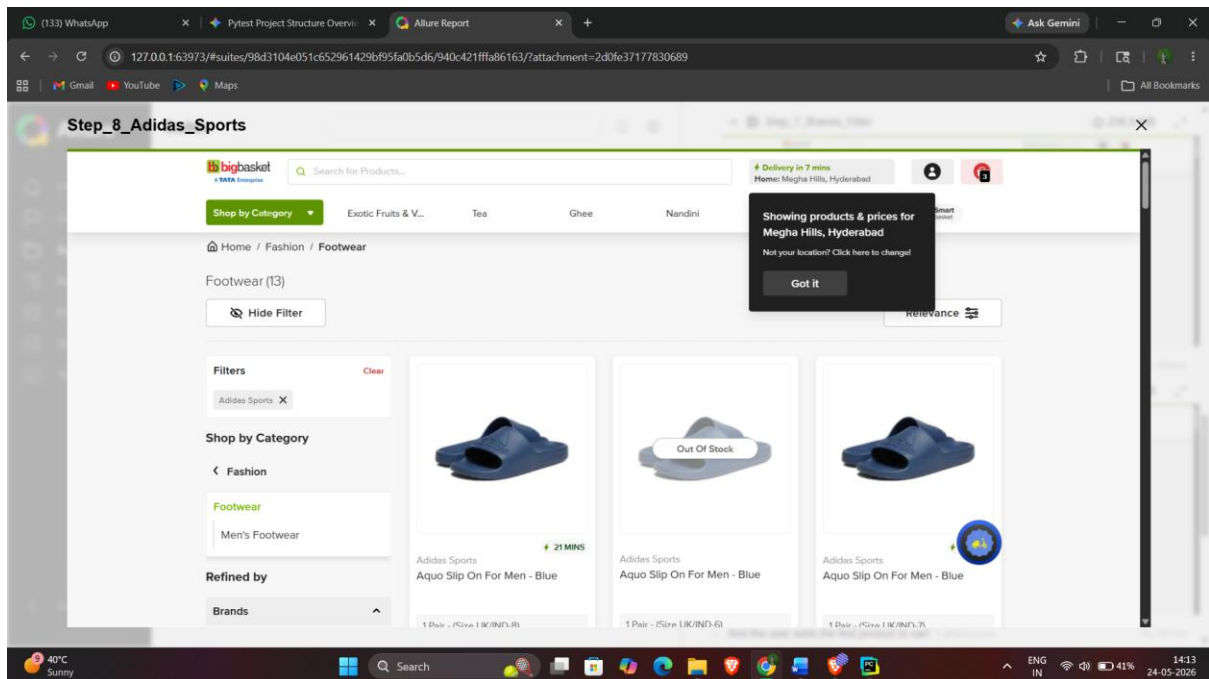
Step 7:-

And the user opens Brands filter



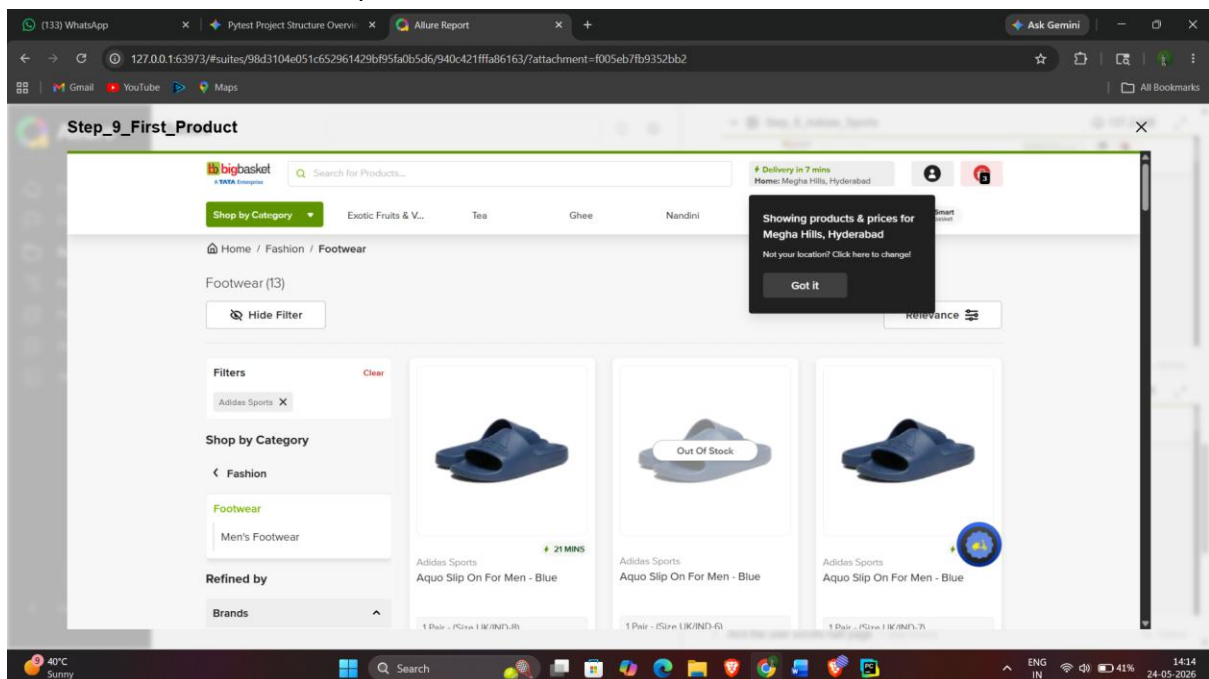
Step 8:-

And the user selects Adidas Sports



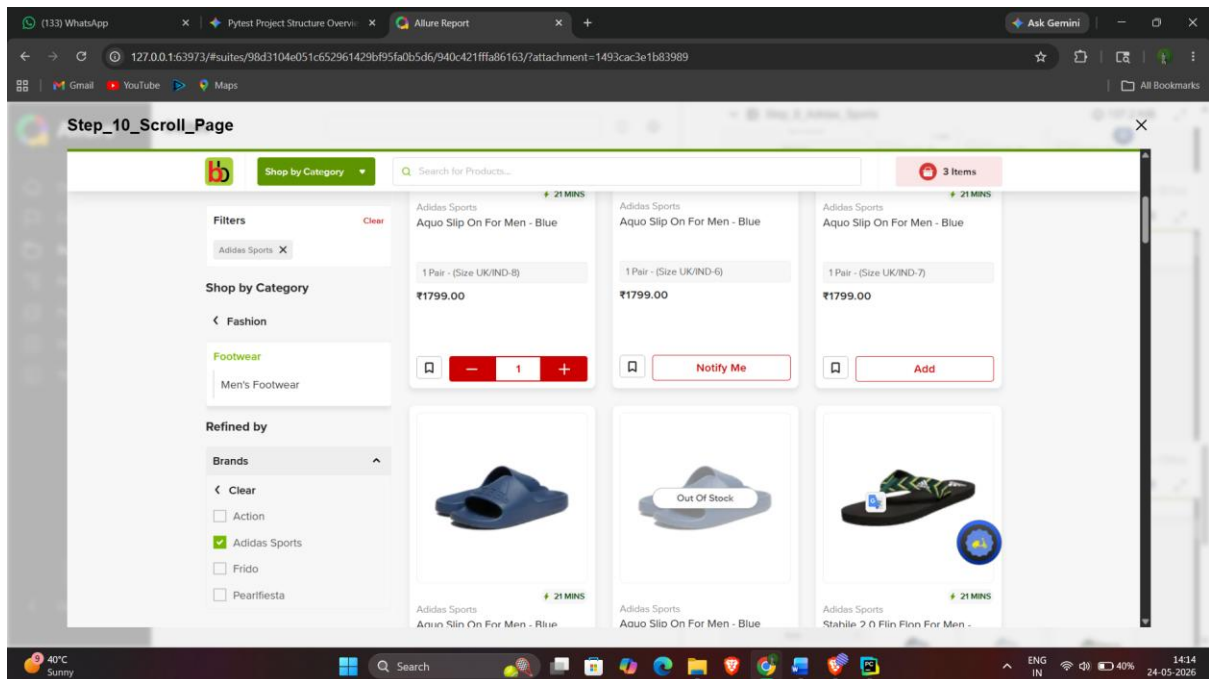
Step 9:-

And the user adds the first product to cart



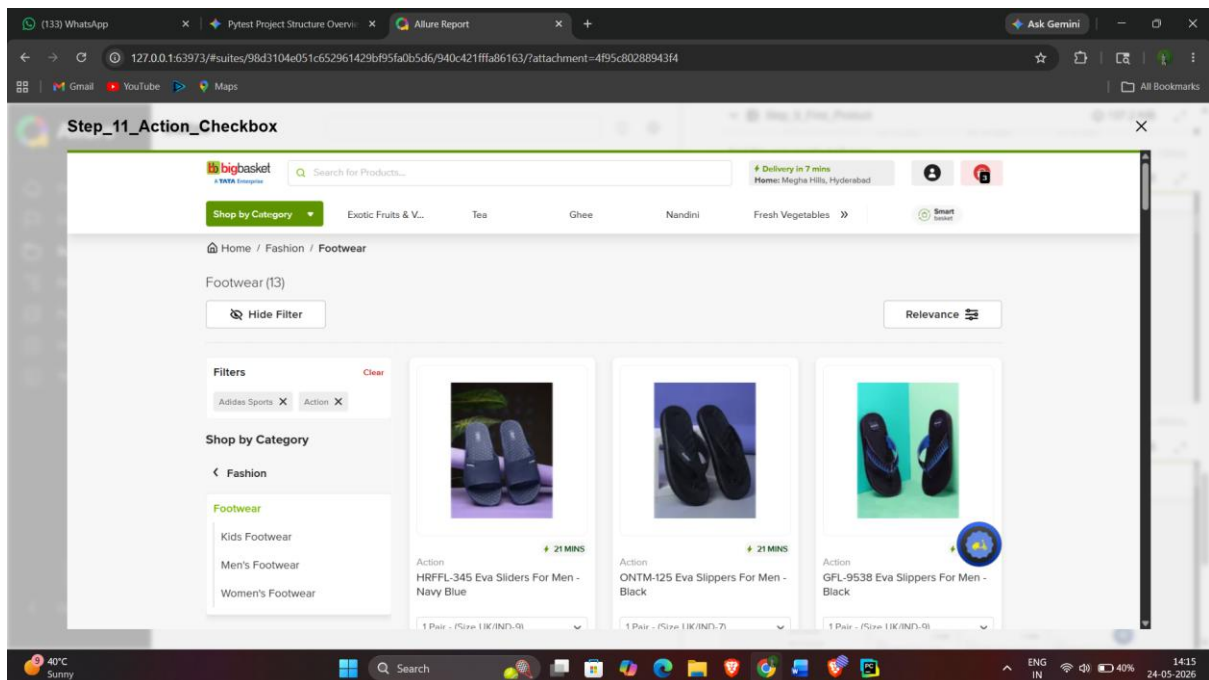
Step 10:-

And the user scrolls half page



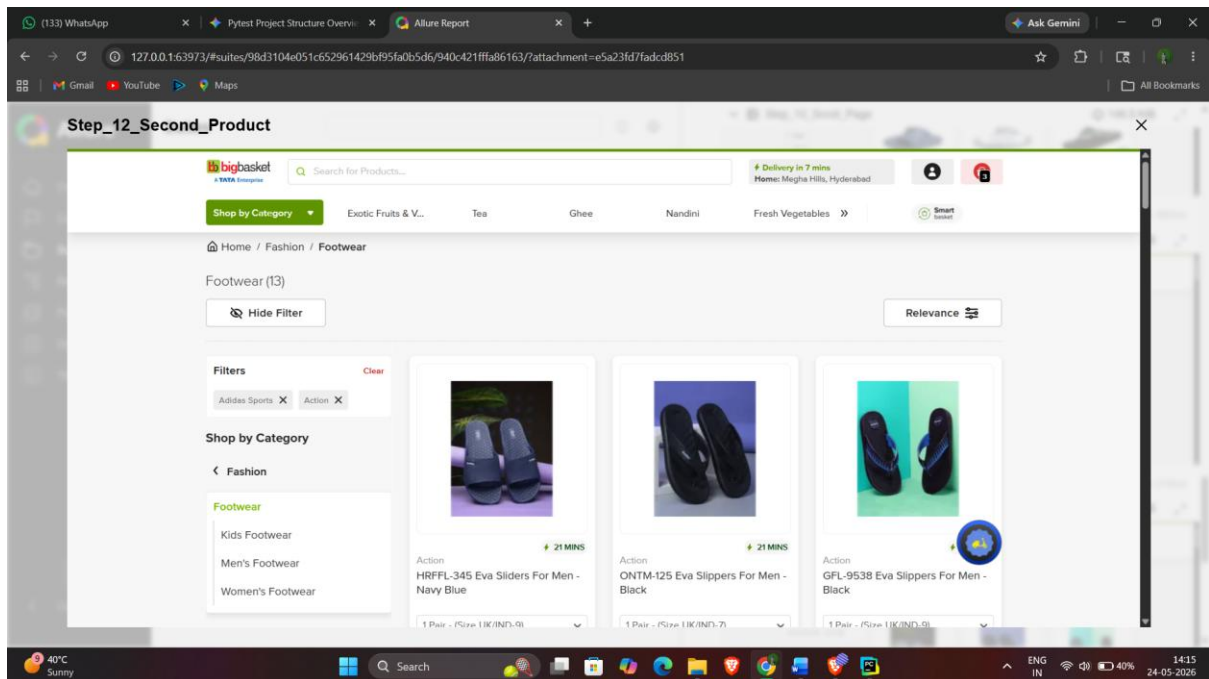
Step 11:-

And the user selects Action checkbox



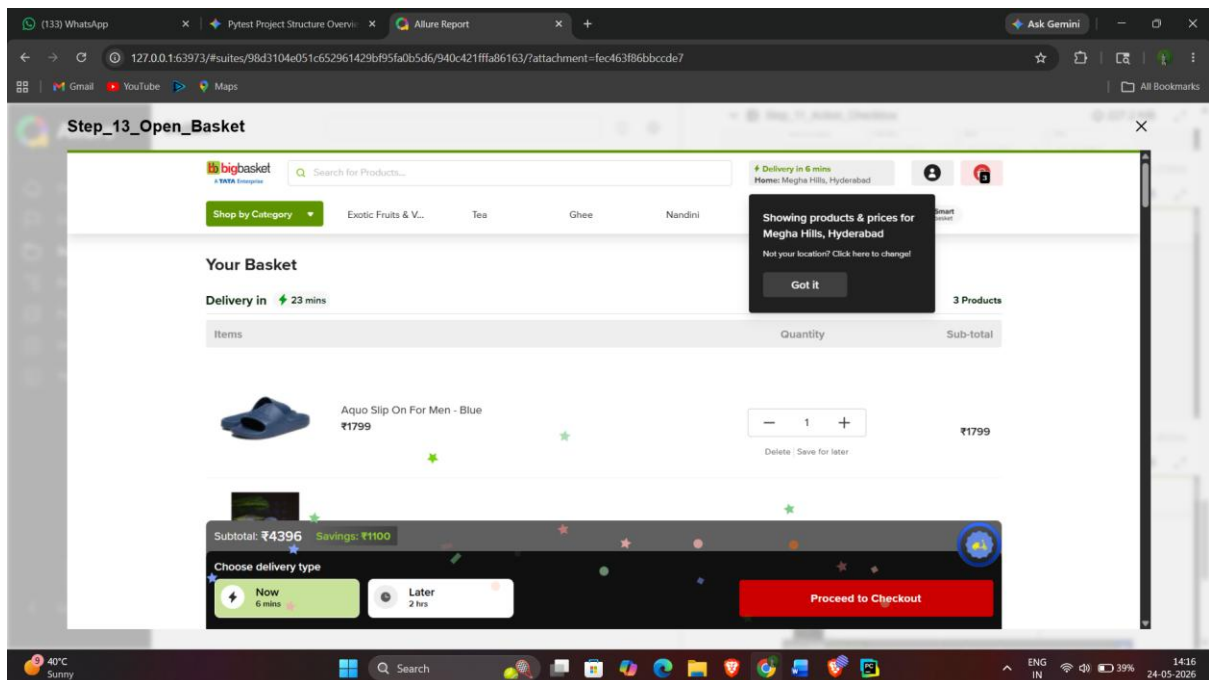
Step 12:-

And the user adds the second product to cart



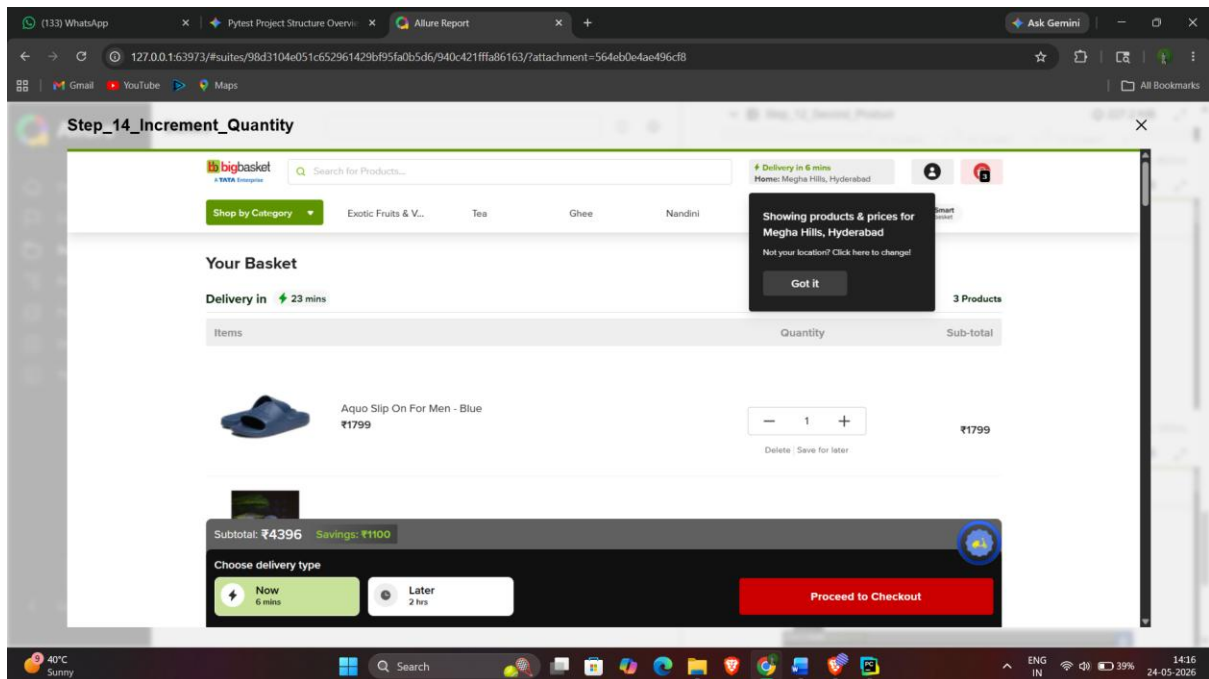
Step 13:-

And the user opens basket with two items



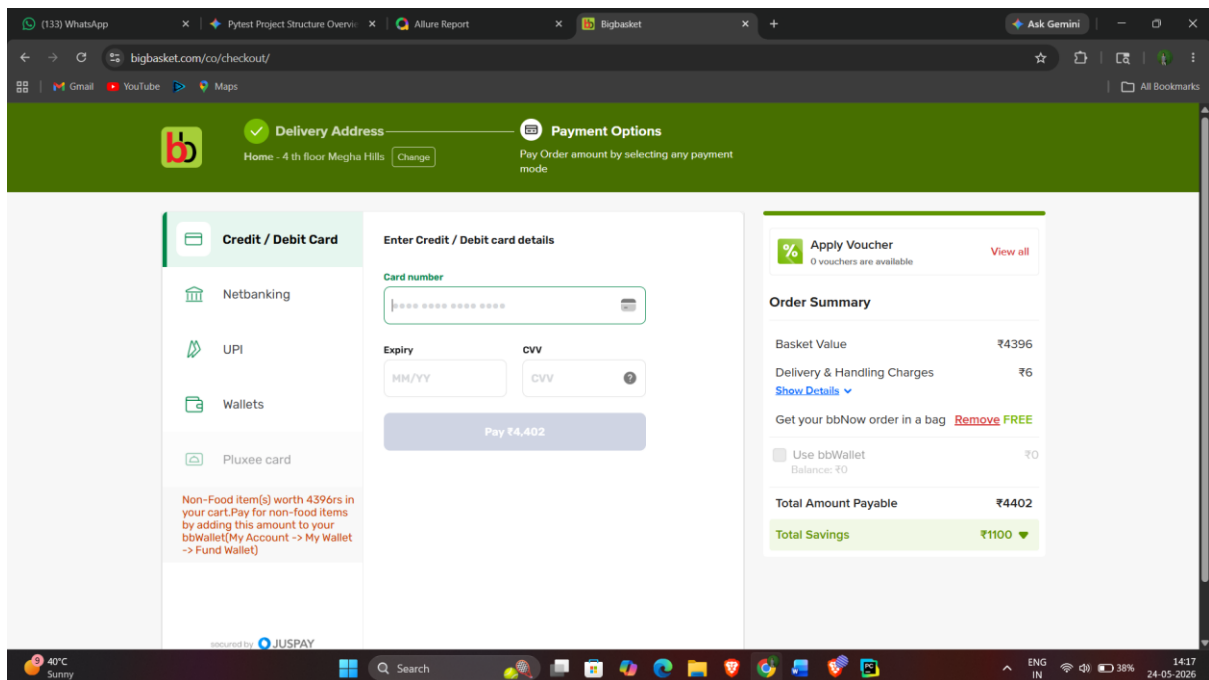
Step 14:-

And the user increments product quantity



Step 15:-

And the user proceeds to checkout

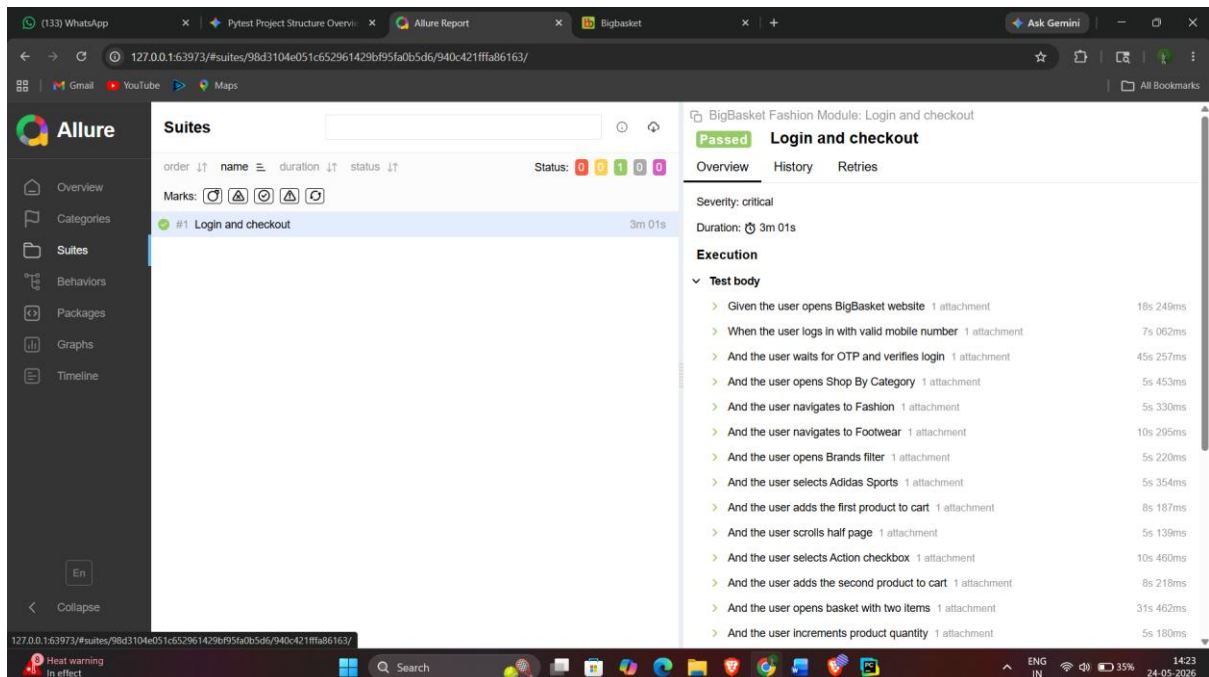
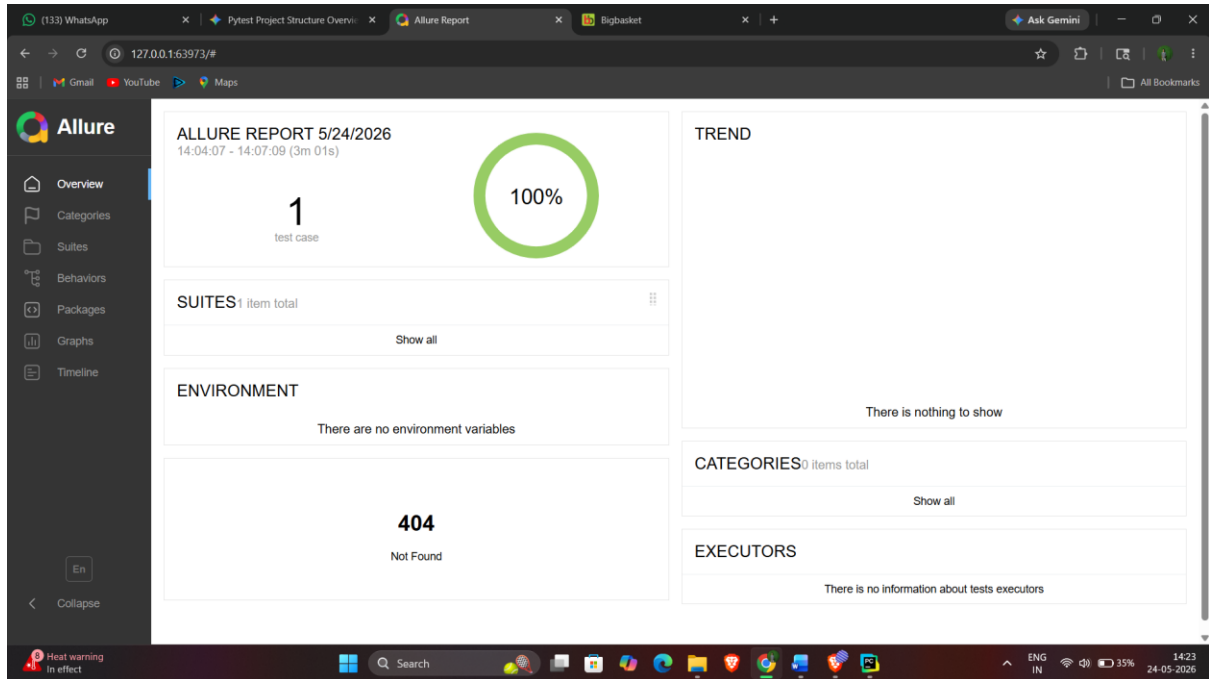


LOGS:

2026-05-21 17:50:58,389 : INFO : Opening BigBasket website
2026-05-21 17:51:01,648 : INFO : BigBasket website opened successfully
2026-05-21 17:51:01,650 : INFO : Entering mobile number
2026-05-21 17:51:20,753 : INFO : Mobile number entered successfully
2026-05-21 17:51:20,755 : INFO : Waiting for OTP entry (manual step)
2026-05-21 17:51:51,586 : INFO : OTP verified and continue clicked
2026-05-21 17:51:51,588 : INFO : Clicking Fashion category
2026-05-21 17:51:57,390 : INFO : Fashion category opened
2026-05-21 17:51:57,392 : INFO : Selecting Mens category
2026-05-21 17:52:04,071 : INFO : Mens category selected
2026-05-21 17:52:04,080 : INFO : Filtering products by Puma brand
2026-05-21 17:52:04,081 : INFO : Adding first product to cart
2026-05-21 17:52:04,128 : INFO : First product added
2026-05-21 17:52:04,129 : INFO : Adding second product to cart
2026-05-21 17:52:04,185 : INFO : Second product added
2026-05-21 17:52:04,186 : INFO : Opening cart page
2026-05-21 17:52:11,473 : INFO : Cart page opened
2026-05-21 17:52:11,482 : INFO : Validating cart page URL
2026-05-21 17:52:11,644 : INFO : Cart page validation passed
2026-05-21 17:52:11,655 : INFO : Clicking proceed to checkout
2026-05-21 17:52:24,761 : INFO : Checkout initiated
2026-05-21 18:06:19,056 : INFO : Opening BigBasket website
2026-05-21 18:06:22,475 : INFO : BigBasket website opened successfully
2026-05-21 18:06:22,476 : INFO : Entering mobile number
2026-05-21 18:06:41,618 : INFO : Mobile number entered successfully
2026-05-21 18:06:41,620 : INFO : Waiting for OTP entry (manual step)
2026-05-21 18:07:12,926 : INFO : OTP verified and continue clicked
2026-05-21 18:07:12,927 : INFO : Clicking Fashion category
2026-05-21 18:07:18,590 : INFO : Fashion category opened
2026-05-21 18:07:18,592 : INFO : Selecting Mens category
2026-05-21 18:07:23,763 : INFO : Mens category selected
2026-05-21 18:07:23,765 : INFO : Filtering products by Puma brand
2026-05-21 18:07:23,766 : INFO : Adding first product to cart
2026-05-21 18:07:23,820 : INFO : First product added
2026-05-21 18:07:23,821 : INFO : Adding second product to cart
2026-05-21 18:07:23,859 : INFO : Second product added
2026-05-21 18:07:23,860 : INFO : Opening cart page
2026-05-21 18:07:30,555 : INFO : Cart page opened

2026-05-21 18:07:30,556 : INFO : Validating cart page URL
2026-05-21 18:07:30,561 : INFO : Cart page validation passed
2026-05-21 18:07:30,563 : INFO : Clicking proceed to checkout
2026-05-21 18:07:43,644 : INFO : Checkout initiated
2026-05-21 18:15:48,978 : INFO : Opening BigBasket website
2026-05-21 18:15:53,737 : INFO : Opening BigBasket website
2026-05-21 18:16:00,929 : INFO : Opening BigBasket website
2026-05-21 22:35:18,800 : INFO : Opening BigBasket website
2026-05-21 22:35:21,625 : INFO : BigBasket website opened successfully
2026-05-21 22:35:21,626 : INFO : Entering mobile number
2026-05-21 22:35:40,224 : INFO : Mobile number entered successfully
2026-05-21 22:35:40,227 : INFO : Waiting for OTP entry (manual step)
2026-05-21 22:39:31,046 : INFO : Opening BigBasket website
2026-05-21 22:39:33,119 : INFO : BigBasket website opened successfully
2026-05-21 22:39:33,120 : INFO : Entering mobile number
2026-05-21 22:39:51,759 : INFO : Mobile number entered successfully
2026-05-21 22:39:51,760 : INFO : Waiting for OTP entry (manual step)
2026-05-21 22:40:22,731 : INFO : OTP verified and continue clicked
2026-05-21 22:40:22,734 : INFO : Clicking Fashion category
2026-05-21 22:40:27,561 : INFO : Fashion category opened
2026-05-21 22:40:27,562 : INFO : Selecting Mens category
2026-05-21 22:40:32,257 : INFO : Mens category selected
2026-05-21 22:40:32,263 : INFO : Filtering products by Puma brand
2026-05-21 22:40:32,265 : INFO : Adding first product to cart
2026-05-21 22:40:32,310 : INFO : First product added
2026-05-21 22:40:32,312 : INFO : Adding second product to cart
2026-05-21 22:40:32,354 : INFO : Second product added
2026-05-21 22:40:32,355 : INFO : Opening cart page
2026-05-21 22:40:38,742 : INFO : Cart page opened
2026-05-21 22:40:38,743 : INFO : Validating cart page URL
2026-05-21 22:40:38,797 : INFO : Cart page validation passed
2026-05-21 22:40:38,798 : INFO : Clicking proceed to checkout
2026-05-21 22:40:51,865 : INFO : Checkout initiated

ALLURE REPORT:-



Browser tabs: (133) WhatsApp, Pytest Project Structure Overview, Allure Report, BigBasket, Ask Gemini

Address bar: 127.0.0.1:63973/#suites/98d3104e051c652961429b95fa0b5d6/940c421ffa86163/

Navigation: Overview, Categories, Suites, Behaviors, Packages, Graphs, Timeline

Suites

order	name	duration	status
#1	Login and checkout	3m 01s	Passed

Status: 0 0 1 0 0

Marks: [Icons]

BigBasket Fashion Module: Login and checkout

Passed Login and checkout

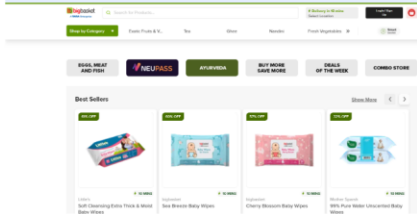
Overview History Retries

Severity: critical
Duration: 3m 01s

Execution

Test body

- Given the user opens BigBasket website 1 attachment 18s 249ms
 - Step_1_Open_BigBasket 265.9 KIB
 - When the user logs in with valid mobile number 1 attachment 7s 062ms
 - And the user waits for OTP and verifies login 1 attachment 45s 257ms
 - And the user opens Shop By Category 1 attachment 5s 453ms



Browser tabs: (133) WhatsApp, Pytest Project Structure Overview, Allure Report, BigBasket, Ask Gemini

Address bar: 127.0.0.1:63973/#graph

Navigation: Overview, Categories, Suites, Behaviors, Packages, Graphs, Timeline

STATUS

100%

Legend: Failed (Red), Broken (Orange), Passed (Green), Skipped (Grey), Unknown (Purple)

SEVERITY

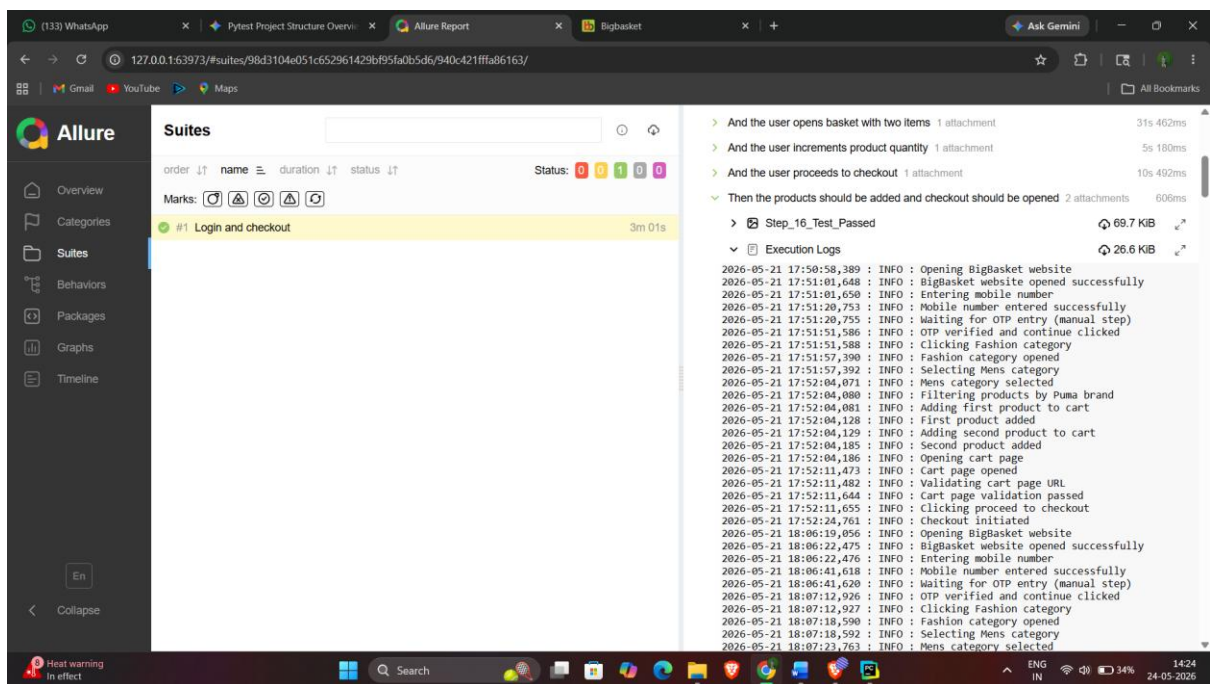
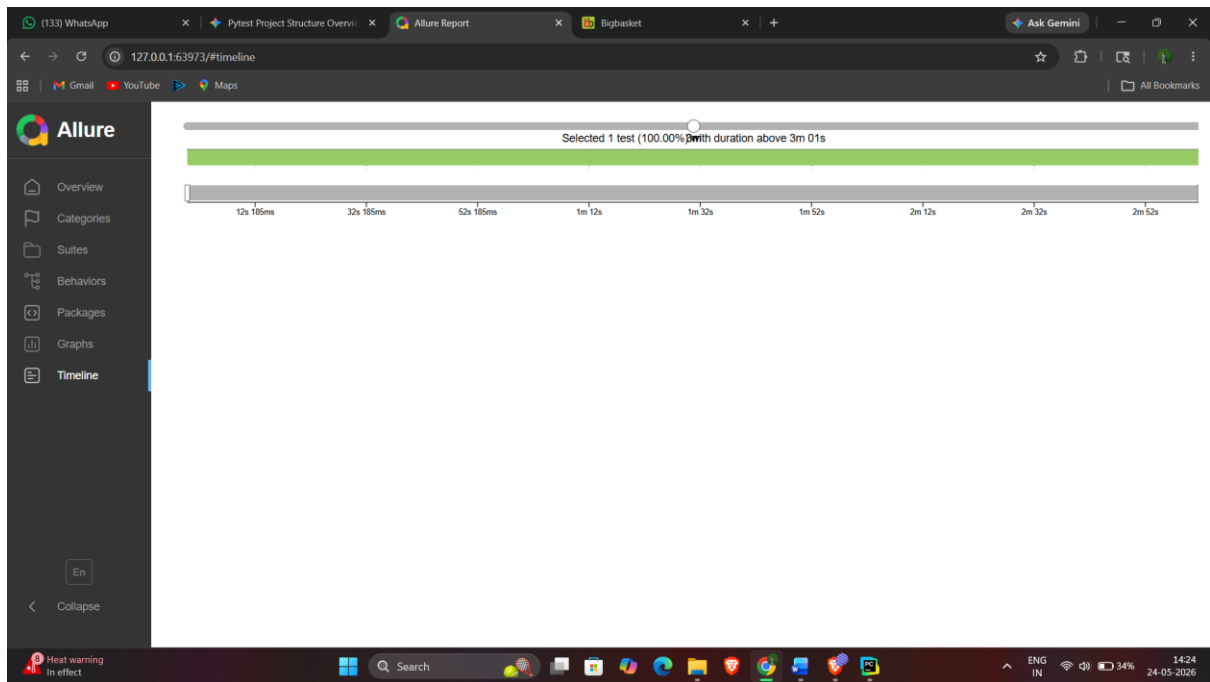
Bar chart showing severity distribution: critical (1), normal, minor, trivial.

DURATION

Bar chart showing duration distribution: 1 (blue bar).

DURATION TREND

Line chart showing duration trend over time.



BDD Functional Testing Flow

Total Tests Executed : 9

Positive Test Cases : 4

Negative Test Cases : 4

End-To-End Scenarios : 1

Step 1: Create Feature Files

1. Test scenarios are written in Gherkin language using:

- o Given
- o When
- o Then

Step 2: Create Step Definitions

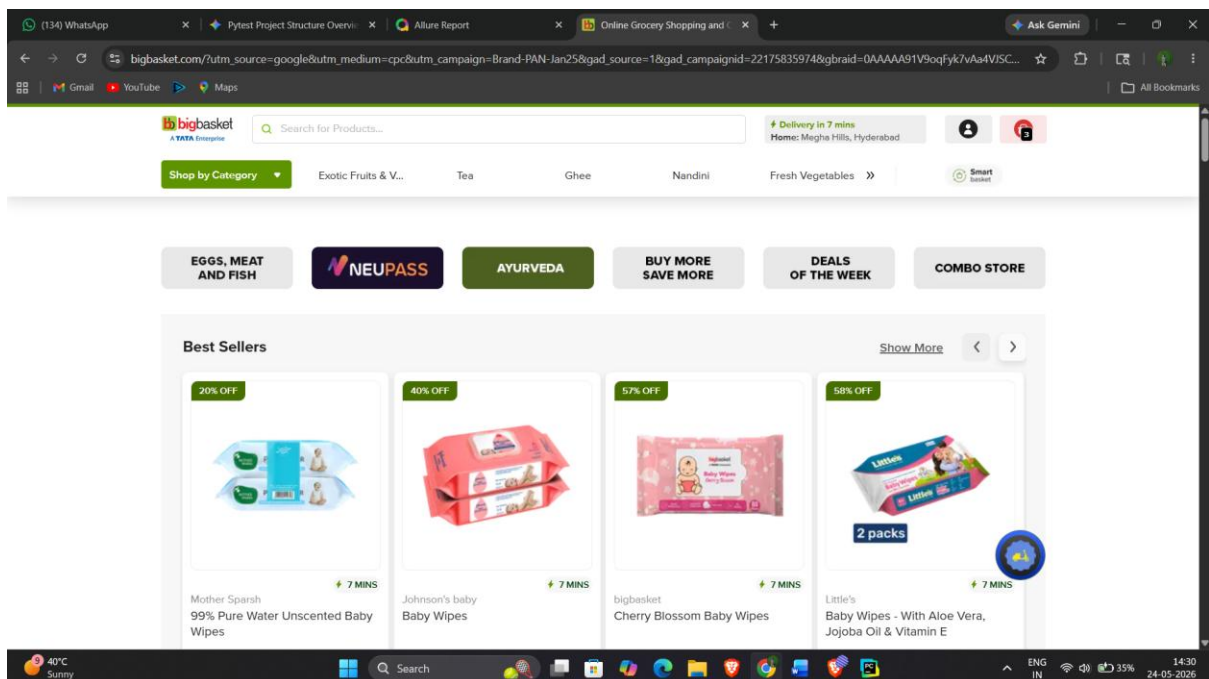
2. Step implementation is written using Behave framework.

Step 3: Launch Browser

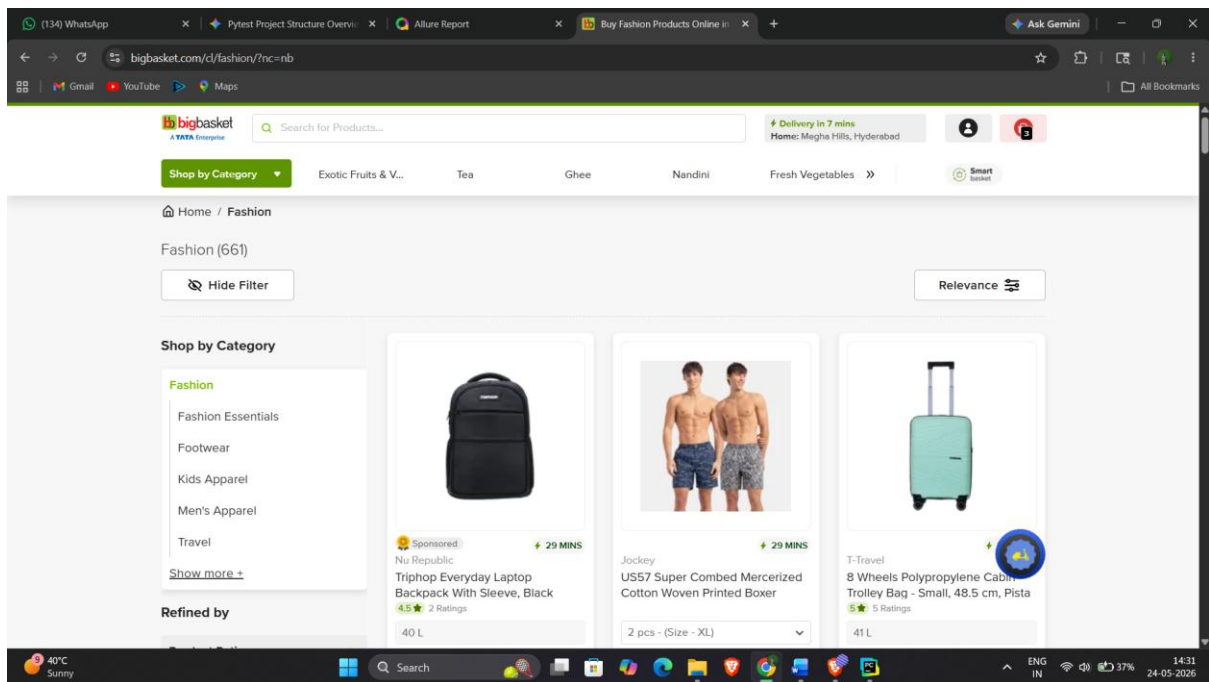
3. Selenium WebDriver launches Chrome browser.

Step 4: Execute Login Scenario

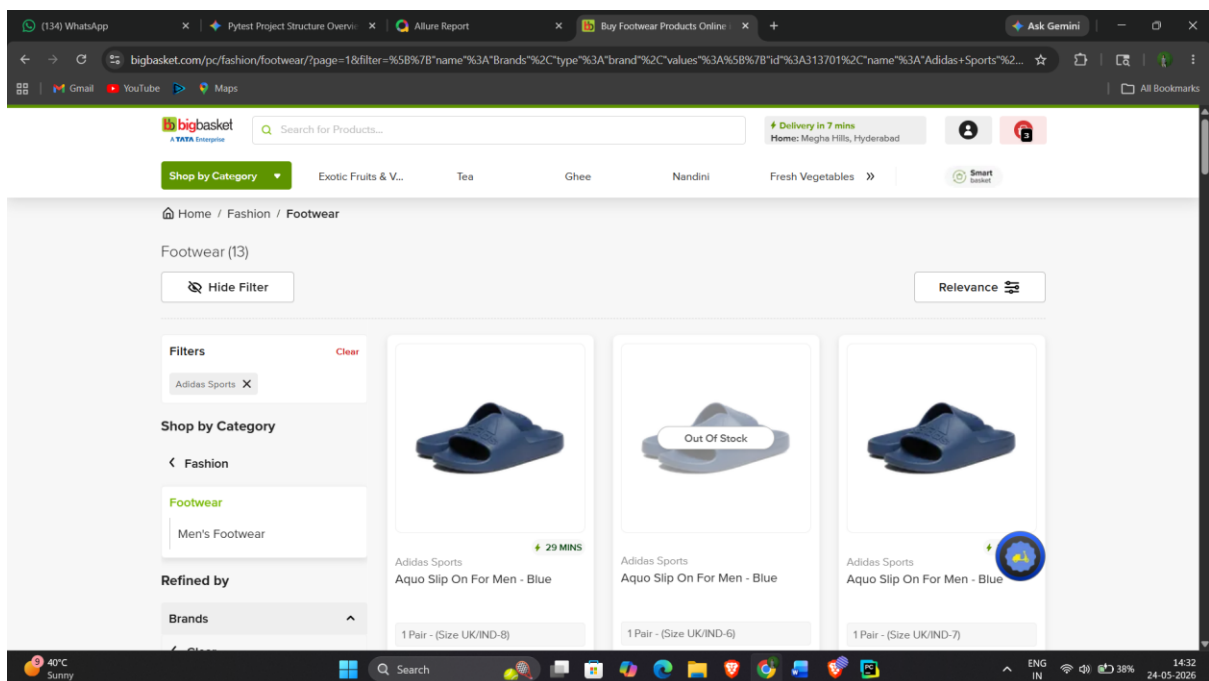
4. User login flow is automated and validated.



Step 5: Execute Product Scenarios

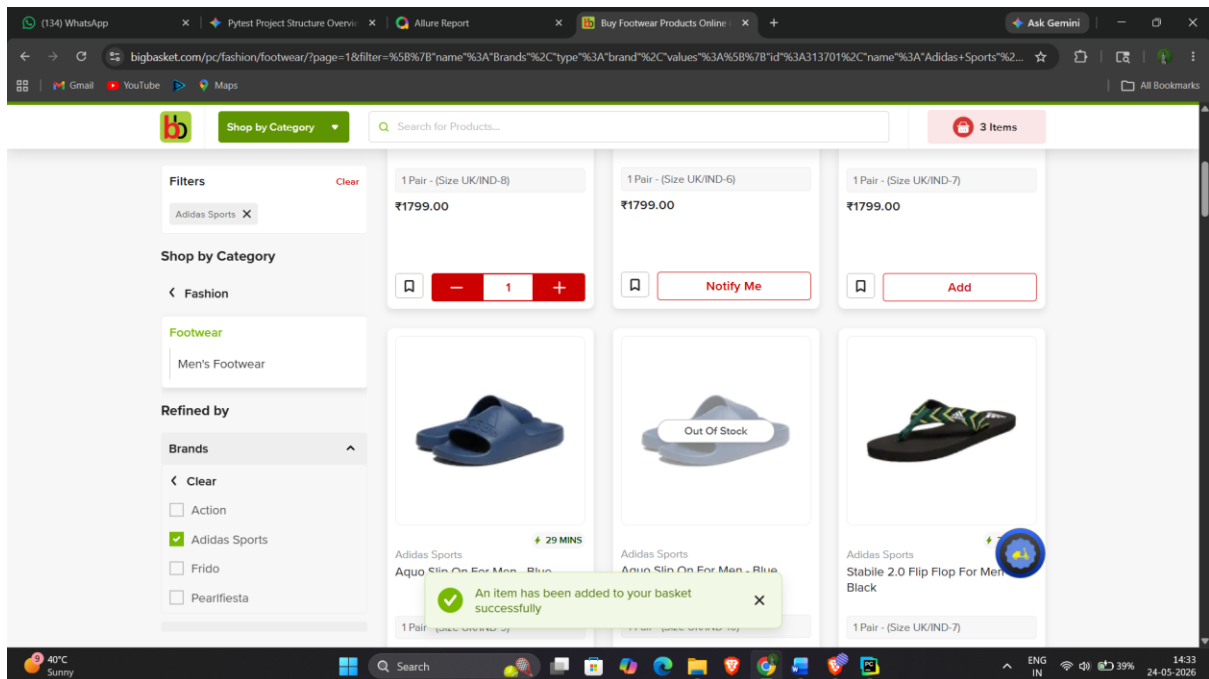


Step 6:- Brand filter validation



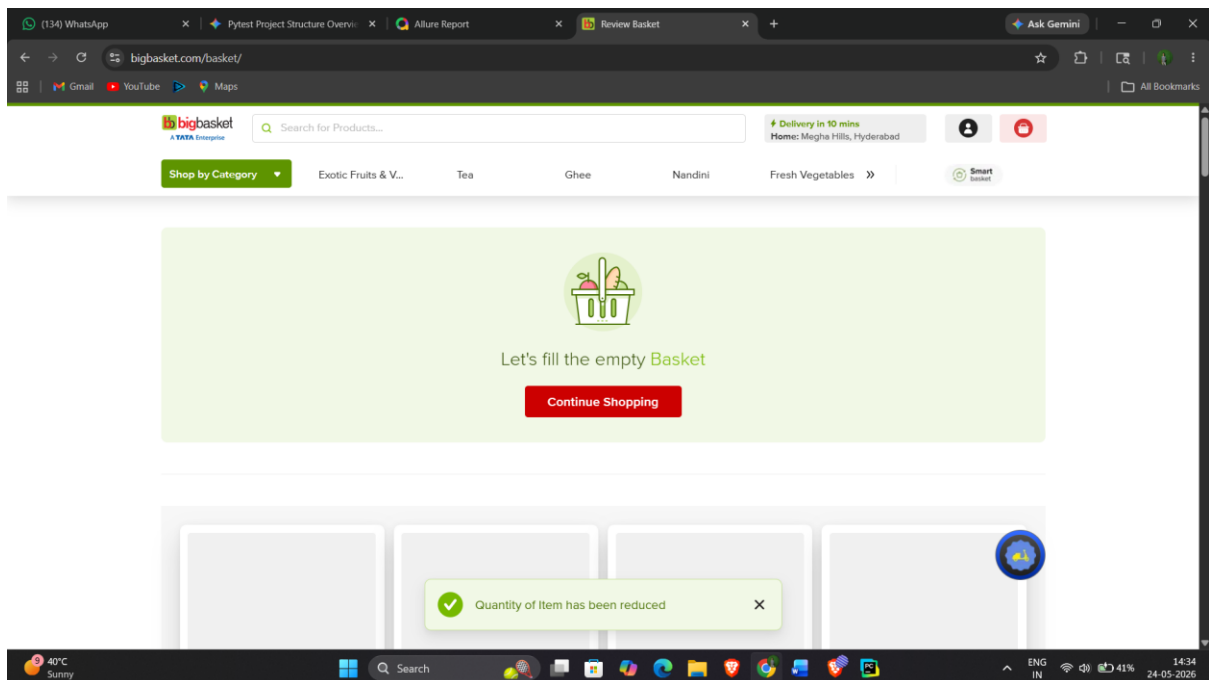
Step 7:-

Add product to cart



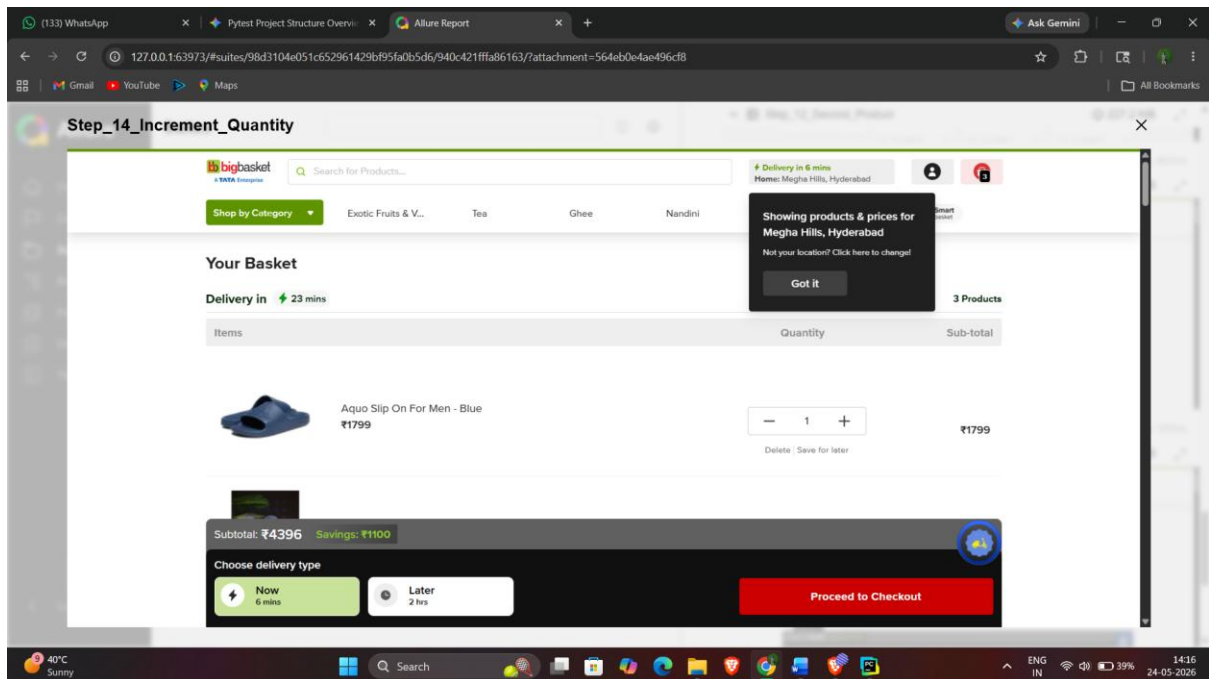
Step 8:-

Delete product from cart



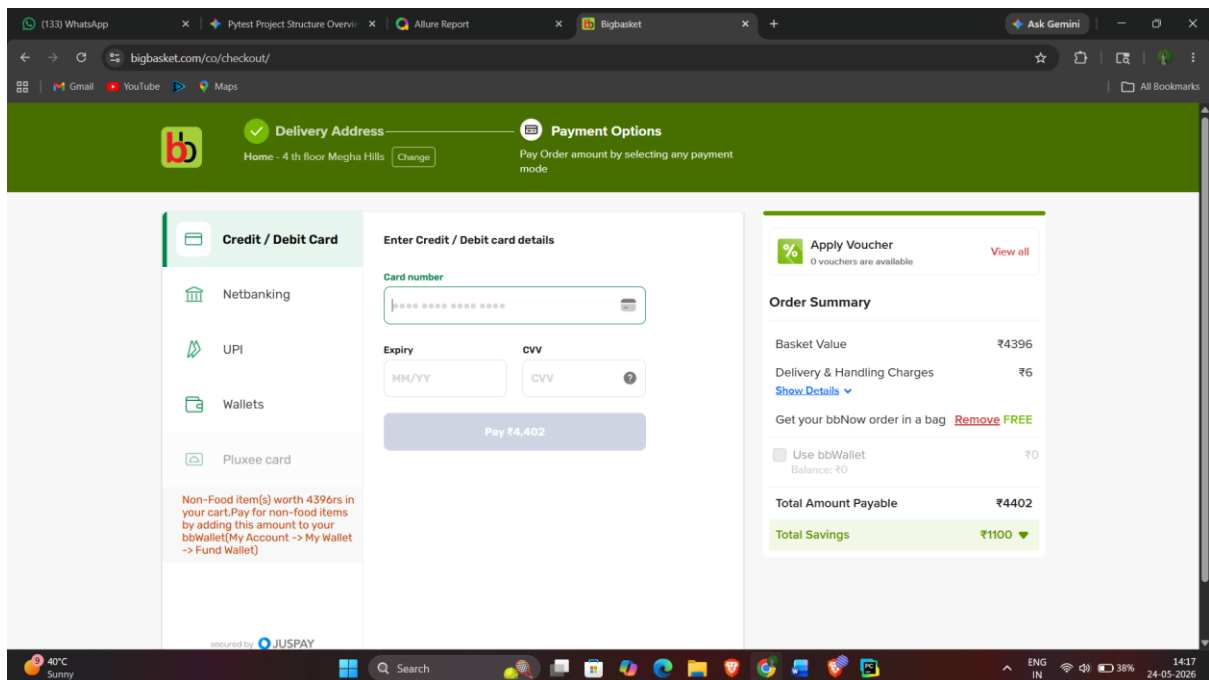
Step 9:-

Execute Checkout Scenarios



Step 10:-

Proceed to checkout

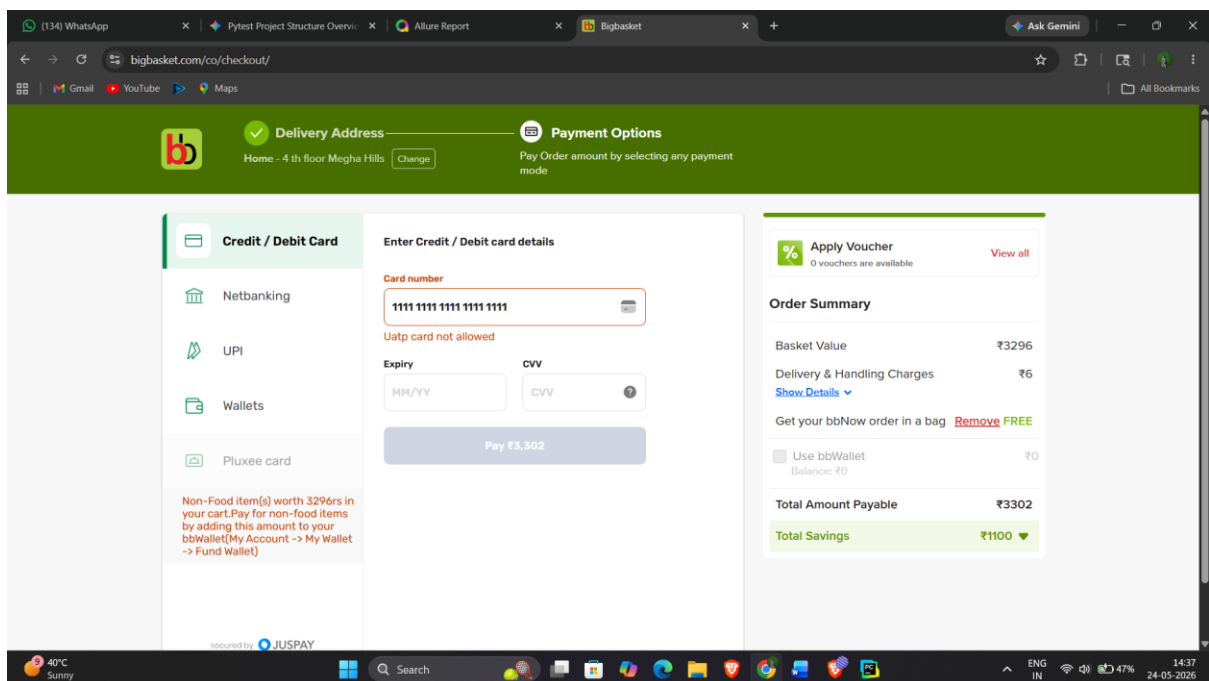


Payment validation

Step 11:-

Execute End-To-End Flow

- 1.Login
- 2.Product selection
3. Cart validation
- 4.Checkout process
- 5.Invalid card validation



Step 12:-

Assertion Validation

1. Assertions validate:
 - o URLs
 - o Product visibility
 - o Error messages
 - o Checkout flow

Step 13:-

- 1.Screenshot And Logging

2. Screenshots captured for execution steps.
3. Logs generated during execution

LOGS:-

2026-05-21 17:50:58,389 : INFO : Opening BigBasket website
2026-05-21 17:51:01,648 : INFO : BigBasket website opened successfully
2026-05-21 17:51:01,650 : INFO : Entering mobile number
2026-05-21 17:51:20,753 : INFO : Mobile number entered successfully
2026-05-21 17:51:20,755 : INFO : Waiting for OTP entry (manual step)
2026-05-21 17:51:51,586 : INFO : OTP verified and continue clicked
2026-05-21 17:51:51,588 : INFO : Clicking Fashion category
2026-05-21 17:51:57,390 : INFO : Fashion category opened
2026-05-21 17:51:57,392 : INFO : Selecting Mens category
2026-05-21 17:52:04,071 : INFO : Mens category selected
2026-05-21 17:52:04,080 : INFO : Filtering products by Puma brand
2026-05-21 17:52:04,081 : INFO : Adding first product to cart
2026-05-21 17:52:04,128 : INFO : First product added
2026-05-21 17:52:04,129 : INFO : Adding second product to cart
2026-05-21 17:52:04,185 : INFO : Second product added
2026-05-21 17:52:04,186 : INFO : Opening cart page
2026-05-21 17:52:11,473 : INFO : Cart page opened
2026-05-21 17:52:11,482 : INFO : Validating cart page URL
2026-05-21 17:52:11,644 : INFO : Cart page validation passed
2026-05-21 17:52:11,655 : INFO : Clicking proceed to checkout
2026-05-21 17:52:24,761 : INFO : Checkout initiated
2026-05-21 18:06:19,056 : INFO : Opening BigBasket website
2026-05-21 18:06:22,475 : INFO : BigBasket website opened successfully
2026-05-21 18:06:22,476 : INFO : Entering mobile number
2026-05-21 18:06:41,618 : INFO : Mobile number entered successfully
2026-05-21 18:06:41,620 : INFO : Waiting for OTP entry (manual step)
2026-05-21 18:07:12,926 : INFO : OTP verified and continue clicked
2026-05-21 18:07:12,927 : INFO : Clicking Fashion category
2026-05-21 18:07:18,590 : INFO : Fashion category opened
2026-05-21 18:07:18,592 : INFO : Selecting Mens category
2026-05-21 18:07:23,763 : INFO : Mens category selected
2026-05-21 18:07:23,765 : INFO : Filtering products by Puma brand

2026-05-21 18:07:23,766 : INFO : Adding first product to cart
2026-05-21 18:07:23,820 : INFO : First product added
2026-05-21 18:07:23,821 : INFO : Adding second product to cart
2026-05-21 18:07:23,859 : INFO : Second product added
2026-05-21 18:07:23,860 : INFO : Opening cart page
2026-05-21 18:07:30,555 : INFO : Cart page opened
2026-05-21 18:07:30,556 : INFO : Validating cart page URL
2026-05-21 18:07:30,561 : INFO : Cart page validation passed
2026-05-21 18:07:30,563 : INFO : Clicking proceed to checkout
2026-05-21 18:07:43,644 : INFO : Checkout initiated
2026-05-21 18:15:48,978 : INFO : Opening BigBasket website
2026-05-21 18:15:53,737 : INFO : Opening BigBasket website
2026-05-21 18:16:00,929 : INFO : Opening BigBasket website
2026-05-21 22:35:18,800 : INFO : Opening BigBasket website
2026-05-21 22:35:21,625 : INFO : BigBasket website opened successfully
2026-05-21 22:35:21,626 : INFO : Entering mobile number
2026-05-21 22:35:40,224 : INFO : Mobile number entered successfully
2026-05-21 22:35:40,227 : INFO : Waiting for OTP entry (manual step)
2026-05-21 22:39:31,046 : INFO : Opening BigBasket website
2026-05-21 22:39:33,119 : INFO : BigBasket website opened successfully
2026-05-21 22:39:33,120 : INFO : Entering mobile number
2026-05-21 22:39:51,759 : INFO : Mobile number entered successfully
2026-05-21 22:39:51,760 : INFO : Waiting for OTP entry (manual step)
2026-05-21 22:40:22,731 : INFO : OTP verified and continue clicked
2026-05-21 22:40:22,734 : INFO : Clicking Fashion category
2026-05-21 22:40:27,561 : INFO : Fashion category opened
2026-05-21 22:40:27,562 : INFO : Selecting Mens category
2026-05-21 22:40:32,257 : INFO : Mens category selected
2026-05-21 22:40:32,263 : INFO : Filtering products by Puma brand
2026-05-21 22:40:32,265 : INFO : Adding first product to cart
2026-05-21 22:40:32,310 : INFO : First product added
2026-05-21 22:40:32,312 : INFO : Adding second product to cart
2026-05-21 22:40:32,354 : INFO : Second product added
2026-05-21 22:40:32,355 : INFO : Opening cart page
2026-05-21 22:40:38,742 : INFO : Cart page opened
2026-05-21 22:40:38,743 : INFO : Validating cart page URL
2026-05-21 22:40:38,797 : INFO : Cart page validation passed
2026-05-21 22:40:38,798 : INFO : Clicking proceed to checkout
2026-05-21 22:40:51,865 : INFO : Checkout initiated

TEST REPORT

Selenium With Pytest

The Selenium Functional Testing framework was executed successfully using the Pytest framework integrated with Selenium WebDriver. The framework automated major functionalities of the BigBasket web application including login validation, product search, filtering, cart operations, checkout validation, and payment validation.

Total Test Cases Executed

7 Test Cases

Test Execution Result

1. Passed Test Cases : 7
2. Failed Test Cases : 0
3. Skipped Test Cases : 0

Accuracy Achieved

1. Test Execution Accuracy : 100%

Functional Modules Tested

1. Login Functionality
2. Fashion Product Search
3. Fashion Product Filter Validation
4. Add Product To Cart
5. Fashion Product Add To cart
6. Checkout Validation
7. Invalid Input Validation

Reporting And Validation During execution:

1. Screenshots were captured automatically
2. Logs were generated successfully
3. Allure HTML Reports were generated
4. Assertions validated application behavior

The Selenium with Pytest framework execution completed successfully without failures, and all validations passed as expected.

BDD With Behave

The BDD Automation Testing framework was executed successfully using Behave integrated with Selenium WebDriver. The framework automated both functional and End-To-End scenarios using Gherkin feature files and reusable step definitions.

The framework validated:

- 1.Positive scenarios
2. Negative scenarios
- 3.Complete End-To-End shopping flow

using Behavior Driven Development methodology.

Total Scenarios Executed

- 1.Functional Scenarios : 8
2. End-To-End Scenarios : 1

Total Test Scenarios:- 9 Scenarios

Test Execution Result

- 1.Passed Scenarios : 9
- 2.Failed Scenarios : 0
- 3.Skipped Scenarios : 0

Accuracy Achieved

- 1.Test Execution Accuracy : 100%

Positive Test Scenarios Executed

- 1.Search Fashion Product
- 2.Apply Fashion Brand Filter
- 3.Add Fashion Product To Cart
- 4.Delete Fashion Product
- 5.Sort Fashion Products
- 6.Checkout Navigation
- 7.End-To-End Shopping Flow

Negative Test Scenarios Executed

- 1.Invalid Mobile Number Validation
- 2.Invalid OTP Validation
- 3.Invalid Product Search
- 4.Invalid Card Validation

End-To-End Testing Flow Executed

The End-To-End testing module successfully automated:

- 1.User Login
- 2.Fashion Product Addition
- 3.Brand Product Addition
- 4.Cart Validation
- 5.Checkout Navigation
- 6.Invalid Payment Validation

Reporting And Validation

The framework successfully generated:

- 1.Allure HTML Reports
- 2.Execution Logs
- 3.Step-Level Screenshots
- 4.Assertion Validation Reports

All scenarios executed successfully and the framework validated the complete functionality of the BigBasket web application without failure

CONCLUSION

The BigBasket Automation Testing Framework was successfully developed and implemented using Selenium WebDriver with Pytest and BDD using Behave. The framework automated major functionalities of the BigBasket web application including login validation, product search, product filtering, cart operations, checkout process, and payment validation.

The project successfully demonstrated:

1. Functional Testing
2. Negative Testing
3. End-To-End Testing
4. Data-Driven Testing
5. Assertion Validation
6. Reporting And Logging

The framework was designed using the Page Object Model (POM) architecture, which improved code reusability, scalability, readability, and maintainability. Integration of Allure Reports, screenshots, and logging mechanisms provided professional reporting and easier debugging capabilities.

Using Pytest improved test execution management and assertion handling, while Behave enabled readable BDD scenarios using Gherkin syntax. The End-To-End testing module validated the complete user workflow from login to checkout process successfully.

All automated test cases and scenarios executed successfully with high accuracy, validating the stability and functionality of the application. The framework reduced manual testing effort, improved execution efficiency, and provided a scalable automation solution suitable for real-time industry projects.

Overall, the project successfully achieved its objective of developing a reliable, maintainable, and professional automation testing framework for web application testing.

